PROJECT PLATINUM:

MINING THE OUTER RIM

A technically plausible asteroid-mining venture

Target: platinum-group metals (platinum + iridium)

Strategy: mine, concentrate, refine and sell high-value cargo

PITCH DECK · ENGINEERING · COST · BUSINESS





01 / MISSION PROJECT PLATINUM

The objective is not merely to mine an asteroid—it is to deliver a product customers will pay for.

THE BUSINESS PROBLEM

- Launch, propulsion, mining rigs and repairs can cost more than the ore is worth.
- A raw asteroid is not a saleable product: composition, purity and delivery matter.
- A large return of rare metal could depress prices; supply must be phased.

OUR PROPOSED SOLUTION

- Scout first; select a metal-rich, accessible near-Earth asteroid.
- Use a reusable solar-electric tug plus robotic mining and processing modules.
- Return only compact, high-value refined cargo; sell under long-term contracts.

DESIGN PROPOSAL

- Autonomous rendezvous, robotic manipulation and fault-detection are mature.
- Solar-electric propulsion is efficient for cargo but slow; no magic faster-than-light tech.
- Platinum-group-metal grade and market price are uncertain until prospecting.

Mission Success Criteria :

1. TECHNICAL

It achieves 99.9% purity grade processed in microgravity environment and maintain 99% autonomous uptime rate.Also includes TRL advancements.

2. ECONOMIC

The extracted metal bring much more profit than it's cost of extraction. Important is that business must be in profit even though metal price drops.

3. COMMERCIAL

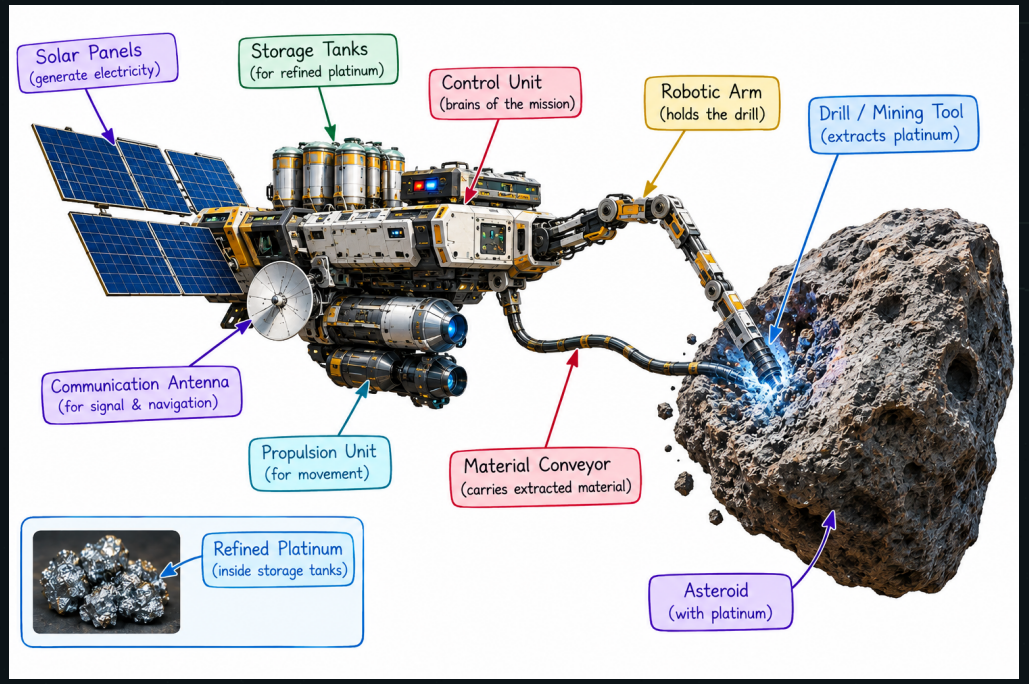
Secure buyers before scaling production.Establish joint ventures with major aerospace manufacturers and filing core patents.

4. ASSURANCE

Mission completes in 3 years with less than 5% failure rate with leaving no orbital waste.Brings 100% Refined Cargo safely back to earth.

02 / ENGINEERING ARCHITECTURE

MINING PROCESS: locate → approach → secure → extract → refine → package → deliver



1 TRANSFER & NAV

Solar-electric tug; Uses star tracker, lidar and optical navigation.

2 ANCHOR & SAMPLE

Secure and test:Harpoons / gripping feet; spectrometer and drill-core sampling.

3 EXCAVATE

Replaceable rotary-percussive drill; enclosed collection hood.

4 PROCESS & REFINE

Sort ore: Crush, sort, separate and concentrate PGMs from raw rocks.

5 PACKAGE & RETURN

Secure cargo: Seal high-value concentrate in shielded, traceable capsules.

Spacecraft Controls:

POWER & CONTROL

- Deployable solar arrays power tools and processing.
- Battery buffers peak drill loads; safe-mode preserves spacecraft.

COMMUNICATION & AUTONOMY OF SPACESHIP

- Deep-space link + local mesh between robots.
- AI plans tasks; Earth approves high-risk actions; store-and-forward telemetry.

FAILURE TOLERANCE

- Redundant avionics, modular tools, spare actuators and self-diagnostics.
- If a drill jams, isolate it and switch to a spare tool.

03 / MINING OPERATION: PLATINUM FROM ROCK

Platinum and iridium are target products; the actual asteroid grade must be verified before committing to production.

A LOCATION

Execute remote sensing + close imaging:

Run Spectrometer, neutron/gamma survey

B CORE SAMPLING

Drill and extract small cores

Measure grade, grain size, volatiles

C ORE EXTRACTION

Anchor securely to the asteroid surface

Cut and capture regolith/rock

D BENEFICIATE

Crush , mill and classify extracted materials

Magnetic / density separation where useful

E METALLURGICAL REFINING

Concentrate PGMs

Smelt or chemical refining in sealed module

F ASSAY + PACKAGING

Test purity and mass

Seal, label and track each capsule

WHY NOT SHIP RAW ASTEROID ROCK?

- Most cargo mass may be worthless host rock.
- Concentration reduces propulsion and capsule costs per unit of valuable metal.
- Do not assume every asteroid contains economically recoverable PGMs.

ENERGY & THERMAL CONTROL

- Solar arrays provide continuous power where illumination permits.
- Battery storage supports high-power drilling and brief shadow periods.
- Radiators reject waste heat; tools pause if temperatures exceed limits.

ENGINEERING BOTTLENECK + BREAKTHROUGH

- Bottleneck: anchoring and handling loose rock in microgravity.
- Breakthrough assumption: autonomous multi-point anchoring + force-feedback tools.
- Validate with a robotic demo before scaling.

Return Decision: **prefer space-based use if customers exist there; Earth return is reserved for small, high-value refined batches.**

04 / COST MODEL: DESIGN FOR MINIMUM COST

Analysis of core cost drivers, strategic mitigations and operational assumptions .All figures reflect architectural planning logic rather than definitive financial forecast

COST DRIVER	WHY IT IS EXPENSIVE	OUR DESIGN CHOICE	EXPECTED EFFECT / ASSUMPTION
1. Launch + propulsion	Massive deep-space payloads and extended transit durations inflate launch costs.	Reusable Solar-Electric Tug: Utilize rideshare launches and optimize trajectory mechanics.	Reduced Propellant Mass: Slower transit profiles maximize efficiency for cargo (non-time-critical payloads).
2. Mining hardware + repairs	Bespoke machinery prone to mechanical failure requires costly replacement launches.	Modular & Redundant Systems: Deploy robotic platforms with swappable drill heads and unified interfaces.	Localized Maintenance: Lower bespoke component footprints allow targeted module replacement over full spacecraft abandonment.
3. Processing + delivery	Transporting raw asteroid material introduces dead-weight penalties from unrefined waste	In-Situ Beneficiation: Concentrate ore at the target site; return only high-value, PGM-rich capsules.	minimized Return Mass: Payload yields become viable once target purity thresholds clear break-even economic margins. .

CONVENTIONAL APPROACH

• Single-use spacecraft; ship bulk ore; ground-heavy processing; limited autonomy.

PROPOSED APPROACH

• Reusable tug + modular robots; process near asteroid; return only high-value product.

Economic gate: Revenue = saleable mass × contract price.

Proceed only when expected revenue exceeds mission + processing + insurance + financing costs, with contingency.

05 / BUSINESS MODEL:MARKETING PROJECT PLATINUM

Build the market before the mine. Start with small batches and contracted buyers, not speculative bulk sales.

WHAT WE SELL

- Primary: assayed platinum-rich concentrate / refined platinum.
- Secondary: iridium-rich fraction if recoverable at useful purity.
- Product sold by certified mass, grade and chain-of-custody.

CUSTOMERS

- Fuel-cell and hydrogen technology suppliers (PGMs as catalysts).
- Electronics / specialist industrial users requiring high-purity metals.
- Potential future in-space manufacturers—only after a real market forms.

WHERE IS IT USED / SOLD?

- A. Earth: established market, but high return and re-entry costs.
- B. In space: avoids Earth return, but demand is currently uncertain.
- C. Lunar / orbital customers: only if local manufacturing scales.

REVENUE MODEL & GO-TO-MARKET

1 PRE-SELL

Letters of intent → minimum-volume offtake contract.

2 DEMONSTRATE

Return a small, independently assayed sample.

3 SCALE GRADUALLY

Add mining robots only after repeatable recovery.

4 MARGIN

Earn on product value minus full lifecycle cost.

Pricing assumption: do not assume today's Earth price stays constant. Cap yearly output, use multi-year contracts and diversify buyers to reduce oversupply risk.

06 / RISKS, VIABILITY & DECISION GATES

A credible investor pitch explains what could fail—and how the team will stop losses early.

TECHNICAL RISK

- Anchor slips, drill jams, robot damage, communication outage.
- Mitigation: redundant sensors, tool spares, autonomy, safe mode and Earth-in-the-loop approvals.
- Gate: complete a simulated excavation and recovery test.

RESOURCE / MARKET RISK

- Asteroid grade or recoverable fraction is lower than expected.
- Market price falls if too much metal is sold at once.
- Mitigation: prospect first, assay samples, phased sales and multiple buyers.

FINANCIAL / SCHEDULE RISK

- Launch delay, cost overrun, insurance limits or long mission duration.
- Mitigation: contingency budget, milestone funding and no full-scale build before demo.
- Gate: independent cost review + signed buyer commitments.

PHASED ROADMAP

PHASE 1 · SCOUT

Remote characterization + target shortlist

PHASE 2 · DEMO

Rendezvous, sampling and small extraction test

PHASE 3 · PILOT

Process a limited batch; verify recovery and purity

PHASE 4 · COMMERCIAL

Repeat missions under signed contracts

Investor recommendation: fund prospecting + technology demonstration first

Do not approve commercial-scale mining until grade, recovery, delivery cost and buyer demand are measured.

THE OUTER RIM IS THE STRATEGIC REALITY OF SPACE RESOURCES

It is a systems-engineering and market-validation challenge.

SCOUT WITH DATA

MINE FOR MARGIN

BUILD ON DEMAND

THANK YOU